

[DC] Multicultural Learning in Virtual Reality to Promote Global Citizenship Education

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ABSTRACT

Due to the pandemic, we have become more reliant on digital technologies in every aspect of our lives. This has led to an increasing interest in teaching and training students and employees using immersive technologies and increasing multicultural learning and digital literacies in schools. This research-creation Ph.D. study aims to advance the understanding of the impact of interactive virtual reality non-fiction (VRNF) narratives as experiential learning tools to promote global citizenship education and to counter radicalization among high school and university students. The researcher will explore the effect of interactive VRNF narratives on open-mindedness trait and intercultural communication competence as moderators for multicultural learning in VR. This study also seeks to design and evaluate a VRNF narrative based on educators/creators' feedback and contribution. This narrative will reflect a personal journey of multicultural learning in the Canadian context. A mixed method approach will be used: Delphi interview method and quasi-experimental study to examine the effects of interactivity and immersion on narrative transportation and user engagement. Results and recommendations will then be reported for further research and implementation.

Keywords: Multicultural learning, virtual reality, interactive narratives, global citizenship education, open-mindedness, intercultural communication.

Index Terms: J.5 [Computer Applications]: arts and humanities; K.3.1 [Computers and Education]: Computer Uses in Education; K.4.m [Computers and Society]: Miscellaneous

1 INTRODUCTION

The overflow of migrants and refugees around the world makes it essential to understand the global context of human rights, diversity, inclusion, equity, and peace. Global Citizenship Education (GCED) employs these concepts, enriches them, and extends their dimensions. It is one of the key education objectives of UNESCO which defines GCED as: "A framing paradigm which encapsulates how education can develop the knowledge, skills, values and attitudes that learners need to build a more just, peaceful and sustainable world and to thrive as global citizens in the twenty first century" [1]. Several practices have been adopted in different countries to promote GCED and to equip their youth with the necessary knowledge, skills, and values to be global citizens. As one promising direction, the design of digital media technologies as educational tools could lead to a new understanding of the global community. Toward this purpose, this Ph.D. research-creation will investigate how implementing and creating interactive virtual reality non-fiction (VRNF) narratives in classrooms can promote

GCED and whether these narratives can help develop active intercultural communication and open-mindedness among students.

With the shifting paradigms of a globalized world, digital platforms including social media are exploited by terrorists to attract and recruit university students, UNESCO acknowledges this problem and the need to tackle radicalization before students' enrollment in universities [1]. Johnson study findings illustrate that the greatest concern is not the content of the curriculum, but the way it is taught in classrooms. Governments provide curriculum framework and policies that promote critical thinking and tolerance to counter extremist ideology; however, it is unclear to educators in schools and universities how to use these ideals for effective pedagogy. Thus, an educational reform to student-centric and open-minded education is needed [2]. To the best of my knowledge, Only one article tackles the potential of virtual reality in countering extremism by proposing it as a new methodological approach for deradicalization programs [3]. In contrast, this research will investigate the use of VR as an experimental tool to promote openminded education and intercultural communication and to help tackle prejudices and radicalized concepts in schools and encouraging students to be global citizens.

VR technologies have already started penetrating educational research. Aside from improving knowledge retention and interest in the curriculum, VR also addresses educational goals (creativity, constructivist learning, empathy, and emotional intelligence) which are not always met through conventional education technologies. As VR presents the opportunity to have meaningful digital experiences, the students can construct their knowledge and apply their solutions creatively within the medium [4]. It was also found that virtual field trips to promote civic and global engagement and stimulate active learning and offline participation [5]. The heightened level of immersion and presence can trigger changes in students' attitudes and promote empathy, active learning, and intercultural understanding [6], [7].

In a different vein, these digital affordances of increased engagement and impact are facing mixed reactions among researchers between skeptical and supporting claims of its emphatical powers [8]. These mixed reactions can be a result of applying the same dynamics of interactivity in traditional and social media in VR [9]. This research aims to investigate the different modes of interactions in VRNF narratives and their effect on narrative transportation. Accordingly, the research will focus on the VR narratives that requires user engagement to move forward in the narrative as previous research indicates that not all artworks exploit VR unique interactive affordances, and VR 360 immersive journalism videos seem to dominate the current artwork [10].

The interactive VRNF narrative that will be created for the purposes of this study aims to promote an understanding of how different cultures relates to others and there are many similarities that cultures share that go beyond their flags and borders. For example, some countries make similar foods but with different names and spices. With a significant amount of research focusing on the use of virtual reality for corporate training and improving employees' performance skills, more research on the use of VRNF narratives as experiential learning tools in the context of GCED and

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intercultural communication is needed, and the experts/educators input and user responses to the designed creation can fill the gap in this area.

2 THEORY

The involvement of the audience/users with the narrative and their interaction with the media content has a range of meanings that vary based on the media channel in question. Several theories explain the effects of narrativity and interactivity on user's cognition, emotion, and behavior, this research will draw on Green & Brock's Narrative Transportation theory (2000), Green & Jenkins (2014) conceptual model of interactivity effects, and David Kolb's Experiential Learning theory (1984) to guide this research and provide insights to inform the design of the creation and its content. The study would consider variables such as presence, immersion, the need for cognition, user control, comfort with the technology, experience taking, and self-referencing when assessing the effects of the interactive VRNF narrative created for the purposes of this study.

Narrative transportation theory and the model of interactive effects in the context of VR non-game experiences seem to be unexplored and understudied research areas. The researcher found only one dissertation that applies the theory in the context of VR games. In her dissertation, Raffel found that interactive VR narratives to be more transportive and engaging than their two-dimensional counterparts and she provided recommendations for designers to build impactful and meaningful experiences in VR [11]. The current research will build on the results of this study and contribute to the literature on the used theoretical frameworks by investigating non-game interactive VRNF narrative experiences in the context of GCED and designing an interactive narrative.

3 PROPOSED METHODOLOGY

A mixed-method approach will be used to view the research problem and objectives from different perspectives. After conducting an extensive review of the literature and an examination of the current VR artwork available and their modes of interaction, the researcher will utilize the delphi method which involves the opinion of field experts to reach a consensus on what are the new standards that could shape the field based on their expertise. The population will be developers and educators with a vision for VR for multicultural learning in classrooms. Media analysis will then be applied to the developed artwork to ensure the utilization of design tools and interaction modes that reflect these newfound standards and processes. Lastly, a final evaluation of the VR experience will be conducted through a quasi-experimental study to examine the effects of interactivity on narrative transportation, open-mindedness and intercultural communication skills. The targeted users are teenagers and youth (13-25 yrs old). Semi-structured interviews or focus groups can also be used to generate more in-depth qualitative insights and feedback.

4 CONCLUSION

The current research- creation attempts to build on the current research findings to advance the practice and research on VR as an experiential learning tool in the context of GCED and intercultural communication which has a still unexplored potential and remains an understudied area. The researcher is actively engaged in coursework in two academic cohorts: [Digital Arts and Humanities](#) and [NSERC CREATE in Immersive Technologies \(CITECH\)](#) and preparing an annotated bibliography and publications for her comprehensive exam.

The field of VR is becoming highly interdisciplinary, and it is developing in a high-speed with interesting and promising venues of research in computer-human interaction, education, and training.

Participation in the IEEE VR 2023 doctoral consortium will allow me the opportunity to present my research to PhD researchers from different disciplines, receive feedback, and engage in discussions of the practical implications of VR in education. Areas of development include (1) Since this research is drawing on multiple theoretical frameworks, discussions can provide insights on which construct will be more useful to the purposes of this study; (2) research questions still need more development to address all aspects of the proposed research and to make sure that research questions don't scope beyond the proposed methodology.

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REFERENCES

- [1] UNESCO, "Global citizenship education: preparing learners for the challenges of the 21st century - UNESCO Digital Library," 2014. <https://unesdoc.unesco.org/ark:/48223/pf0000227729> (accessed Dec. 12, 2022).
- [2] D. Johnson, "Intelligence and the management of radicalisation and extremism in universities in Asia and Africa," in *The Routledge International Handbook of Universities, Security and Intelligence Studies.*, Routledge, 2019, pp. 326–339. doi: 10.4324/9780203702086-24.
- [3] P. Pelletier and E. Drozda-Senkowska, "Virtual reality as a tool for deradicalizing the terrorist mind: Conceptual and methodological insights from intergroup conflict resolution and perspective-taking research," *Peace Confl. J. Peace Psychol.*, vol. 26, no. 4, pp. 449–459, Nov. 2020, doi: 10.1037/pac0000442.
- [4] H. Au and J. Lee, "Virtual reality in education: a tool for learning in the experience age," *Int. J. Innov. Educ.*, vol. 4, no. 4, pp. 215–226, Apr. 2018, doi: DOI:10.1504/IJIE.2017.10012691.
- [5] S. Delacruz, "Building Digital Literacy Bridges: Connecting Cultures and Promoting Global Citizenship in Elementary Classrooms through School-Based Virtual Field Trips," *TechTrends*, vol. 63, no. 4, pp. 428–439, Jul. 2019, doi: 10.1007/s11528-018-0350-1.
- [6] E. Whewell, H. Caldwell, G. Malloch, and B. Garrett, "Digital Learning Across Boundaries: Immersive technologies supporting changemaking in an international context," *Res. Educ. Learn. Innov. Arch.*, no. 27, Art. no. 27, Jul. 2021, doi: 10.7203/realia.27.18019.
- [7] J. Parong and R. E. Mayer, "Learning science in immersive virtual reality," *J. Educ. Psychol.*, vol. 110, pp. 785–797, 2018, doi: 10.1037/edu0000241.
- [8] T. Gruenewald and S. Witteborn, "Feeling good: humanitarian virtual reality film, emotional style and global citizenship," *Cult. Stud.*, vol. 36, no. 1, pp. 141–161, Jan. 2022, doi: 10.1080/09502386.2020.1761415.
- [9] G. Raz, "Rage against the empathy machine revisited: The ethics of empathy-related affordances of virtual reality," *Convergence*, vol. 28, no. 5, pp. 1457–1475, Oct. 2022, doi: 10.1177/13548565221086406.
- [10] C. Bevan *et al.*, "Behind the Curtain of the 'Ultimate Empathy Machine': On the Composition of Virtual Reality Nonfiction Experiences," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, Glasgow Scotland Uk, May 2019, pp. 1–12. doi: 10.1145/3290605.3300736.
- [11] S. Raffel, "Narrative Transportation and Virtual Reality: Exploring the Immersive Qualities of Social Justice in the Digital World," *Electron. Theses Diss.*, Jan. 2018, [Online]. Available: <https://stars.library.ucf.edu/etd/5800>